

Erfan Amani Bani

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Education

London Business School <i>Ph.D. in Management Science and Operations</i>	London, UK 2025 – 2030 (Expected)
Sharif University of Technology <i>M.Sc. in Industrial Engineering (Optimization Methods)</i>	Tehran, Iran 2023 – 2025
Sharif University of Technology <i>B.Sc. in Industrial Engineering</i>	Tehran, Iran 2019 – 2023

Research Interests

Methodology: Optimization, machine learning.

Applications: Healthcare operations, supply chain and logistics network design.

Publications

- [1] **Amani Bani, E., & Eshghi, K.** (2026). *Quantum modeling of the dynamic ride-sharing problem: Development of quantum Benders decomposition methods*. Expert Systems with Applications, 310, 131320. [DOI]
- [2] Fallahi, A., **Amani Bani, E., & Varmazyar, M.** (2025). *Towards sustainable scheduling of unrelated parallel batch processors: A multi-objective approach with triple bottom line, classical and data-driven robust optimization*. Computers & Operations Research, 173, 106863. [DOI]
- [3] **Amani Bani, E., Fallahi, A., Varmazyar, M., & Fathi, M.** (2022). *Designing a sustainable reverse supply chain network for COVID-19 vaccine waste under uncertainty*. Computers & Industrial Engineering, 174, 108808. [DOI]
- [4] Fallahi, A., **Amani Bani, E., & Niaki, S. T. A.** (2022). *A constrained multi-item EOQ inventory model for reusable items: Reinforcement learning-based differential evolution and particle swarm optimization*. Expert Systems with Applications, 207, 118018. [DOI]

Professional Experience

Snapp! Mar. 2025 – Sep. 2025
Senior AI Developer (Largest ride-hailing and delivery platform in the Middle East) *Tehran, Iran*

- Engineered low-latency AI pipelines for large-scale customer service operations.
- Built classification models to detect multihoming drivers, supporting platform-loyalty analytics.
- Designed ML-based behavioural prediction models informing dynamic incentive policies across a network of 2M+ drivers.

FLYTZI

Co-founder & Optimization Engineer

Feb. 2024 – Feb. 2025

Tehran, Iran

- Co-founded a mid-mile logistics platform matching vendors with drivers for parcel delivery.
- Formulated and solved the underlying capacitated vehicle routing problem with time windows; developed metaheuristic solvers for real-time dispatch at operational scale.
- Designed a 3D bin-packing module for parcel-to-vehicle assignment, integrating geometric feasibility with routing decisions.

Teaching Experience

Sharif University of Technology, Department of Industrial Engineering

2022 – 2025

- **Head Teaching Assistant:** Statistics; Engineering Economics; Plant Layout; Project Control.
- **Teaching Assistant:** Operations Research; Probability Theory; Simulation.

Academic Service

Journal Reviewer: *Transportation Research Part E; Expert Systems with Applications; IEEE Transactions on Pattern Analysis and Machine Intelligence.*

Volunteering

Gamein — Nationwide Gamified Education Event

2022 – 2023

Game Design and Development Team Lead

Sharif University of Technology

- Led product development for a national educational simulation reaching 3,300+ participants across Iranian universities.
- Coordinated a cross-functional team spanning game design, software engineering, and operations modelling.

Honors & Awards

- LBS PhD Fellowship, full funding, London Business School, 2025–2030.
- Direct admission to M.Sc. programme (Talented Student Quota), Sharif University of Technology, 2023.
- Top 0.2% nationally, Iranian University Entrance Exam (Konkur), among 150,000+ candidates, 2019.

Last updated: May, 2026